

To: Editor-in-Chief

Journal: IEEE Transactions on Neural Networks and Learning Systems (TNNLS)

Date: August 23, 2026

Subject: Submission of Research Manuscript: “Diffeomorphic Implicit Fourier Neural Operators for Fast PDE Solving on Arbitrary Irregular Domains”

Dear Editor-in-Chief and Editorial Board,

I am pleased to submit our research article titled “**Diffeomorphic Implicit Fourier Neural Operators for Fast PDE Solving on Arbitrary Irregular Domains**” for consideration for publication as a Regular Paper in *IEEE Transactions on Neural Networks and Learning Systems*.

Core Scientific Contributions:

- **Novel Operator Architecture:** DIF-FNO overcomes the grid-restricted limitations of standard Fourier Neural Operators via learnable implicit diffeomorphic coordinate transformations.
- **Jacobian Regularization:** Metric distortion penalty based on the Jacobian determinant, ensuring smooth mapping to a canonical torus while preventing high-frequency Gibbs boundary artifacts.
- **Empirical Performance:** On complex non-convex star domains, DIF-FNO achieves a Relative L^2 error of 13.51% and Sobolev H^1 error of 13.51% with a compact footprint of $0.90M$ parameters.

Declarations:

- This manuscript is original, has not been published previously, and is not under consideration elsewhere.
- All authors have read and approved the final manuscript.

Sincerely,

Giovanni D’Agnese

Department of Computer Science / Engineering